Part I: Research Question

A1: QUESTION FOR ANALYSIS

Does the payment method have an impact on the customer's Churn rate?

A2: BENEFIT FROM ANALYSIS

From this analysis, the stakeholders in the organization will understand our customers' values on the payment method in determining whether to continue doing business with the company, and such has a high impact on customer retention. Through exploratory data analysis, the stakeholders will also understand if the collected data meets the business needs.

A3: DATA IDENTIFICATION

The following variables were considered in answering the research question: Does the payment method impact the customer's Churn rate?

1. Churn

Churn is used as a dependent variable since it is the most relevant variable in our decision-making process. The churn represents whether the customer has left the business or still with the company.

2. Payment Method

The PaymentMethod variable indicates the customer's payment method. The data type is categorical. The Payment Method is an independent variable. The payment method has 4 levels of cardinality comprising Credit Card (automatic), Bank transfer(automatic), Mailed Check, and electronic check.

B1: CODE

The chi-square test will be used to analyze the Churn dataset.

Code attached

#To run the code, please modify the input path variable in the code

InputFilePath = 'C:/Users/beatr/Downloads/churn_clean.csv'

BeatriceMwangi_D2
07_ExploratoryData/

B2: OUTPUT

Results:

Power_divergenceResult(statistic=439.1336, pvalue=7.371814027426512e-95)

Difference between PaymentMethod is statistically significant

From the output, the payment method is statistically significant in determining the churning of the customer in the telecom industry.

Summarized code output.

```
#B : ANALYSIS OF DATASET USING CHI SQUARE
#Chi- sQuare Test

#Extract Churn Ratio
Churn_Ratio =Churn_df['PaymentMethod'].value_counts()
# Perform Chi-square test
chi= stats.chisquare(Churn_Ratio)
print(chi)

# Test significance
alpha= 0.05
if chi[1] < alpha:
    print("Difference between PaymentMethod is statistically significant")
else:
    print("No significant difference between PaymentMethod found")
```

```
Power_divergenceResult(statistic=439.1336, pvalue=7.371814027426512e-95)
Difference between PaymentMethod is statistically significant
```

B3: JUSTIFICATION

I chose to use the non-parametric chi-square test to analyze the Payment method categorical data since the Chi-square can be utilized with categorical data.

According to McHugh, Mary(2013, January 8), the Chi-square statistic is a non-parametric (distribution-free) tool designed to analyze the group differences when the dependent variable is measured at a nominal level.

C: UNIVARIATE STATISTICS

According to Bobbitt and Zach (2021, November 22), univariate analysis refers to the analysis of one variable.

- **Two continuous variables**

1. MonthlyCharge

The MonthlyCharge variable represents the monthly amount charged to the customer. This value reflects an average per customer. The data type is continuous quantitative. The monthly charge is an independent variable.

2. Bandwidth_GB_Year

The Bandwidth_GB_Year represents the average data the customer uses in GB in a year. The data type is continuous quantitative. The Bandwidth_GB_Year is an independent variable.

- **Two categorical variables**

1. Payment Method

The PaymentMethod variable indicates the customer's payment method. The data type is categorical. The Payment Method is an independent variable. The payment method has 4 levels of cardinality comprising Credit Card (automatic), Bank transfer(automatic), Mailed Check, and electronic check.

2. Area

According to the census data, the Area variable describes the zoning of the customer's areas of residence. The data is categorical and has 3 cardinality levels comprising urban, suburban, and rural categories.

C1: VISUAL OF FINDINGS

Univariate Analysis of Categorical Data

1. **Payment method**

The code

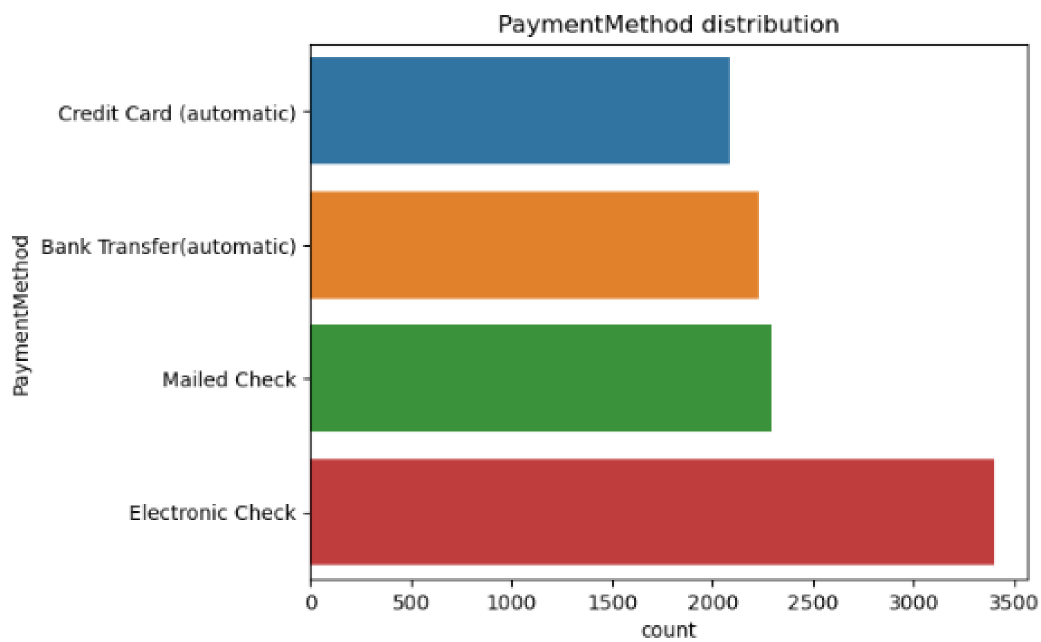
```
print('# UNIVARIATE ANALYSIS:CATEGORICAL VARIABLE 1')
print('')
# C UNIVARIATE ANALYSIS
#CATEGORICAL VARIABLE 1
plt.title('PaymentMethod distribution')
sns.countplot(data=Churn_df, y="PaymentMethod",hue="PaymentMethod")
plt.show()
print(Churn_df[['PaymentMethod']].describe())

print('')
print(Churn_df.groupby('PaymentMethod')['PaymentMethod'].count())
```

The distribution

```
           PaymentMethod
count              10000
unique                 4
top     Electronic Check
freq                3398

PaymentMethod
Bank Transfer(automatic)    2229
Credit Card (automatic)     2083
Electronic Check            3398
Mailed Check                2290
Name: PaymentMethod, dtype: int64
```

The Visual

```
# UNIVARIATE ANALYSIS:CATEGORICAL VARIABLE 1
```




```
          PaymentMethod
count             10000
unique                4
top    Electronic Check
freq               3398

PaymentMethod
Bank Transfer(automatic)    2229
Credit Card (automatic)     2083
Electronic Check            3398
Mailed Check                2290
Name: PaymentMethod, dtype: int64
```

2. Area

The code

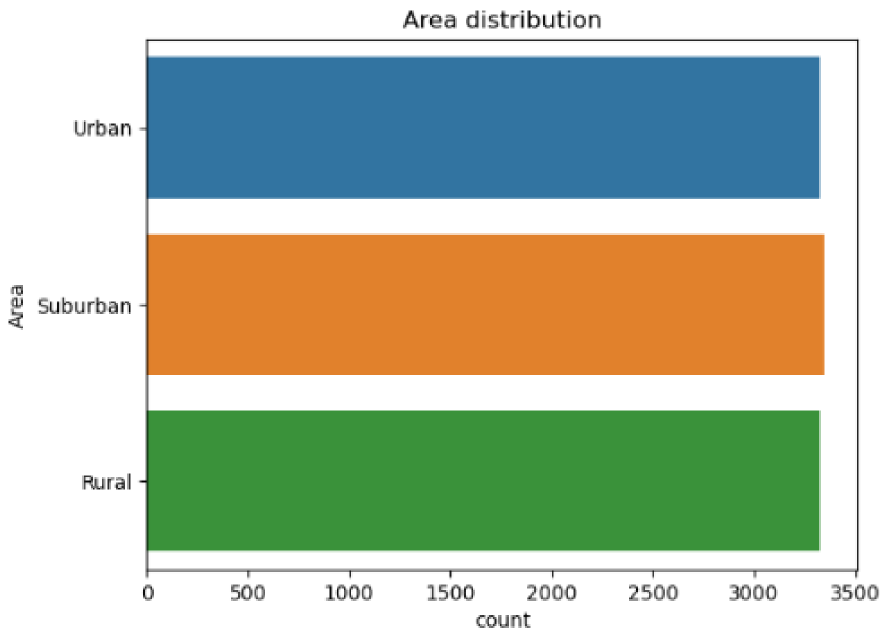
```
print('# UNIVARIATE ANALYSIS:CATEGORICAL VARIABLE 1')
print('')
# C UNIVARIATE ANALYSIS
#CATEGORICAL VARIABLE 2
plt.title('Area distribution')
sns.countplot(data=Churn_df, y="Area",hue="Area")
plt.show()
#churn_plot.show()
print('Area distribution')
print(Churn_df[['Area']].describe())
print('')
print(Churn_df.groupby('Area')['Area'].count())
```

The distribution

```
Area distribution
            Area
count      10000
unique         3
top     Suburban
freq        3346

Area
Rural       3327
Suburban    3346
Urban       3327
Name: Area, dtype: int64
```

The Visual

```
# UNIVARIATE ANALYSIS:CATEGORICAL VARIABLE 1
```




```
Area distribution
            Area
count      10000
unique         3
top     Suburban
freq        3346

Area
Rural       3327
Suburban    3346
Urban       3327
Name: Area, dtype: int64
```

Univariate analysis of continuous data

1. Monthly Charge

The code

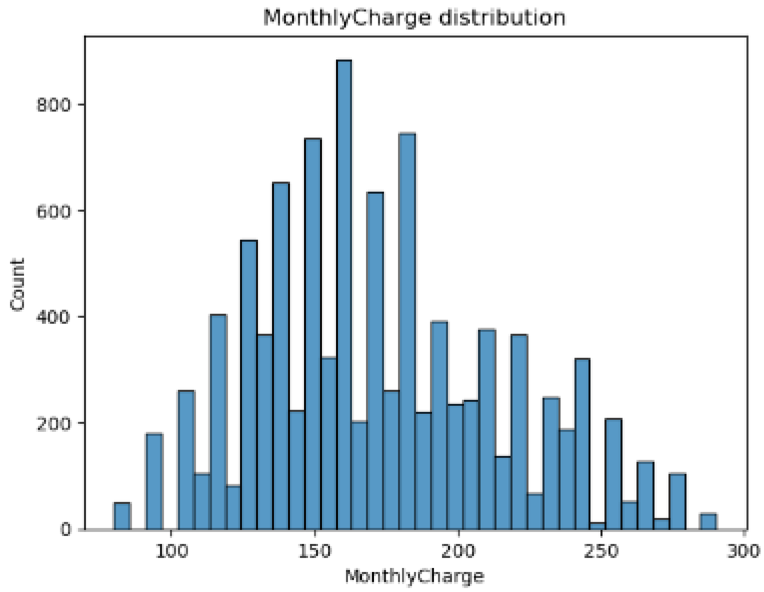
```
print('# UNIVARIATE ANALYSIS:CONTINOUS VARIABLE 1')
print('')

# C UNIVARIATE ANALYSIS
#CONTINOUS VARIABLE 1
plt.title('MonthlyCharge distribution')
MonthlyChargeDF =Churn_df[['MonthlyCharge']]

sns.histplot(MonthlyChargeDF,x='MonthlyCharge')

plt.show()
plt.title('MonthlyCharge distribution')
sns.boxplot( data=Churn_df, x='MonthlyCharge')
plt.show()
print( Churn_df['MonthlyCharge'].describe())
```

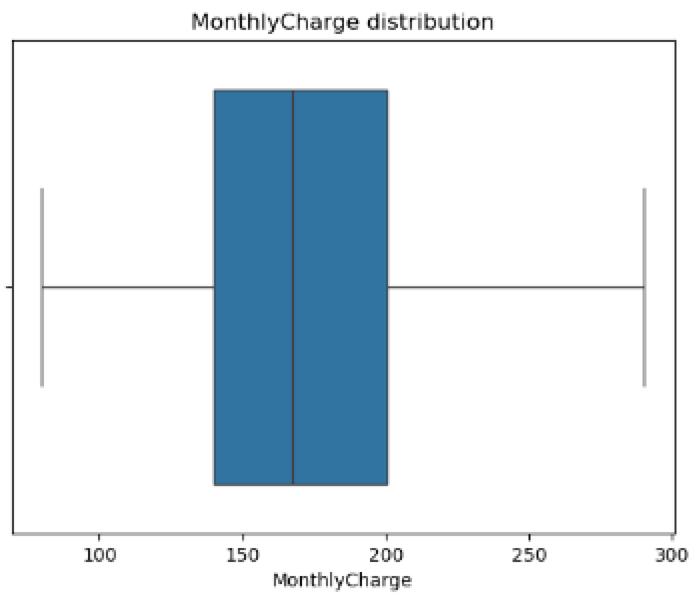
The distribution

```
count    10000.000000
mean       172.624816
std         42.943094
min         79.978860
25%        139.979239
50%        167.484700
75%        200.734725
max        290.160419
Name: MonthlyCharge, dtype: float64
```

The Visual

```
# UNIVARIATE ANALYSIS:CONTINOUS VARIABLE 1
```






```
count    10000.000000
mean       172.624816
std         42.943094
min         79.978860
25%        139.979239
50%        167.484700
75%        200.734725
max        290.160419
Name: MonthlyCharge, dtype: float64
```

2. Bandwidth_GB_Year

The code

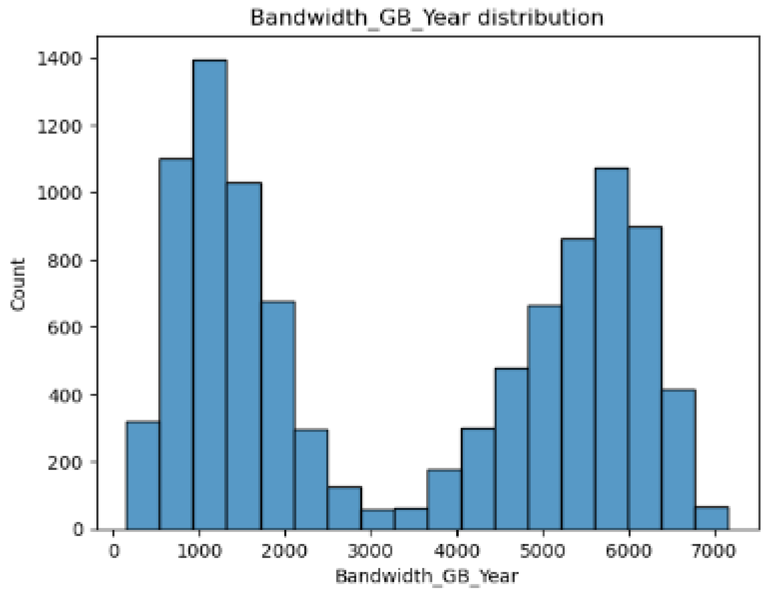
```
print('# UNIVARIATE ANALYSIS:CONTINOUS VARIABLE 2')
print('')

# C UNIVARIATE ANALYSIS
#CONTINOUS VARIABLE 1
plt.title('Bandwidth_GB_Year distribution')
TenureDF =Churn_df[['Bandwidth_GB_Year']]

sns.histplot(TenureDF,x='Bandwidth_GB_Year')

plt.show()
plt.title('Bandwidth_GB_Year distribution')
sns.boxplot( data=Churn_df, x='Bandwidth_GB_Year')
plt.show()
print( Churn_df['Bandwidth_GB_Year'].describe())
```

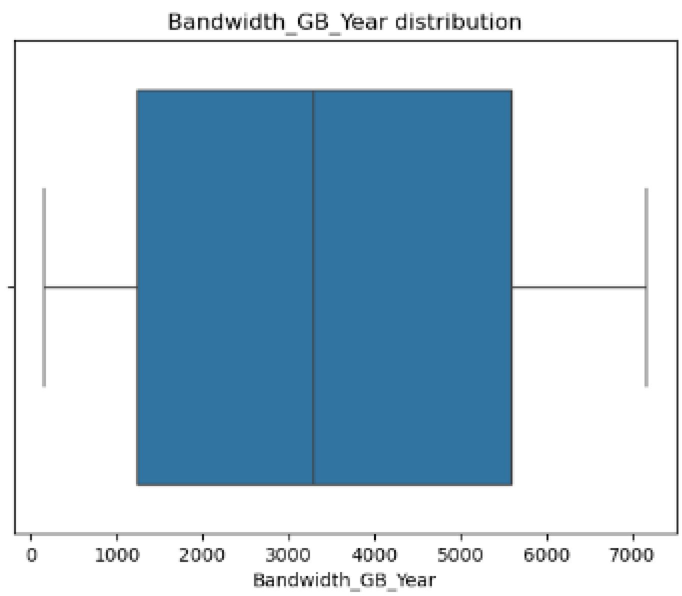
The distribution

```
count    10000.000000
mean      3392.341550
std       2185.294852
min        155.506715
25%       1236.470827
50%       3279.536903
75%       5586.141370
max       7158.981530
Name: Bandwidth_GB_Year, dtype: float64
```

The Visual

```
# UNIVARIATE ANALYSIS:CONTINOUS VARIABLE 2
```






```
count    10000.000000
mean      3392.341550
std       2185.294852
min        155.506715
25%       1236.470827
50%       3279.536903
75%       5586.141370
max       7158.981530
Name: Bandwidth_GB_Year, dtype: float64
```

D: BIVARIATE STATISTICS

According to Bobbitt, Zach (2021, November 22), bivariate analysis refers to the analysis of two variables, and the purpose of bivariate analysis is to understand the relationship between two variables.

According to Ashwini(2023, November 27), Bivariate analysis is a systematic statistical technique applied to a pair of variables (features/attributes) to establish their empirical relationship.

- **Two continuous variables**

1. Bandwidth_GB_Year

   The Bandwidth_GB_Year represents the average data the customer uses in GB in a year. The data type is continuous quantitative. The Bandwidth_GB_Year is an independent variable. The Bandwidth_GB_Year will be analyzed against the Tenure variable, which indicates the number of months the customer has stayed to establish their relationship.

2. Monthly Charge

   The MonthlyCharge variable represents the monthly amount charged to the customer. This value reflects an average per customer. The data type is continuous quantitative. The monthly charge is an independent variable. The Monthly charge will be analyzed against the income to establish if there is any relationship between the two variables.

- **Two categorical variables**

1. Payment Method

The PaymentMethod variable indicates the customer's payment method. The data type is categorical. The Payment Method is an independent variable. The payment method has 4 levels of cardinality comprising Credit Card (automatic), Bank transfer(automatic), Mailed

Check, and electronic check. The payment method will be analyzed against the Churn to determine the relationship between the variables.

2. Area

According to the census data, the Area variable describes the zoning of the customer's areas of residence. The data is categorical and has 3 cardinality levels comprising urban, suburban, and rural categories. The Area will be analyzed against the Churn to determine the relationship between the variables.

D1: VISUAL OF FINDINGS

1. Bandwidth_GB_Year vs Tenure

The Code

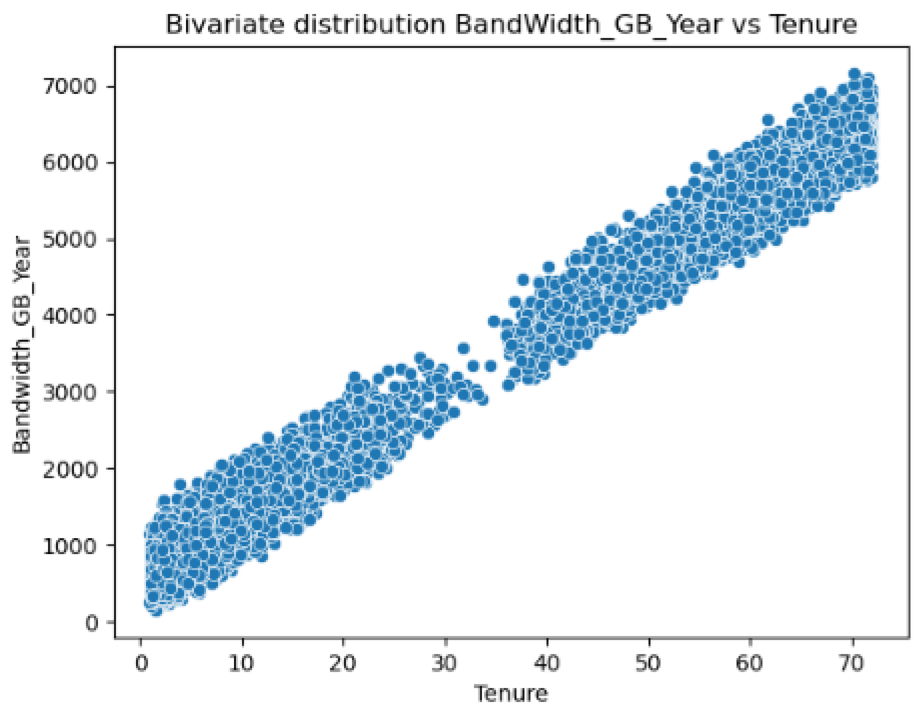
```
# BIVARIATE ANALYSIS
#CONTINOUS VARIABLE 1
print('# BIVARIATE ANALYSIS:CONTINOUS VARIABLE 1')


plt.title('Bivariate distribution BandWidth_GB_Year vs Tenure')
sns.scatterplot(data=Churn_df, x='Tenure', y='Bandwidth_GB_Year')
plt.show()
print(Churn_df[['Tenure','Bandwidth_GB_Year']].describe())
```

The Distribution

	Tenure	Bandwidth_GB_Year
count	10000.000000	10000.000000
mean	34.526188	3392.341550
std	26.443063	2185.294852
min	1.000259	155.506715
25%	7.917694	1236.470827
50%	35.430507	3279.536903
75%	61.479795	5586.141370
max	71.999280	7158.981530

The Visual

```
# BIVARIATE ANALYSIS:CONTINOUS VARIABLE 1
```




```
          Tenure  Bandwidth_GB_Year
count  10000.000000       10000.000000
mean      34.526188        3392.341550
std       26.443063        2185.294852
min        1.000259         155.506715
25%        7.917694        1236.470827
50%       35.430507        3279.536903
75%       61.479795        5586.141370
max       71.999280        7158.981530
```

2. Monthly Charge vs Income

The Code

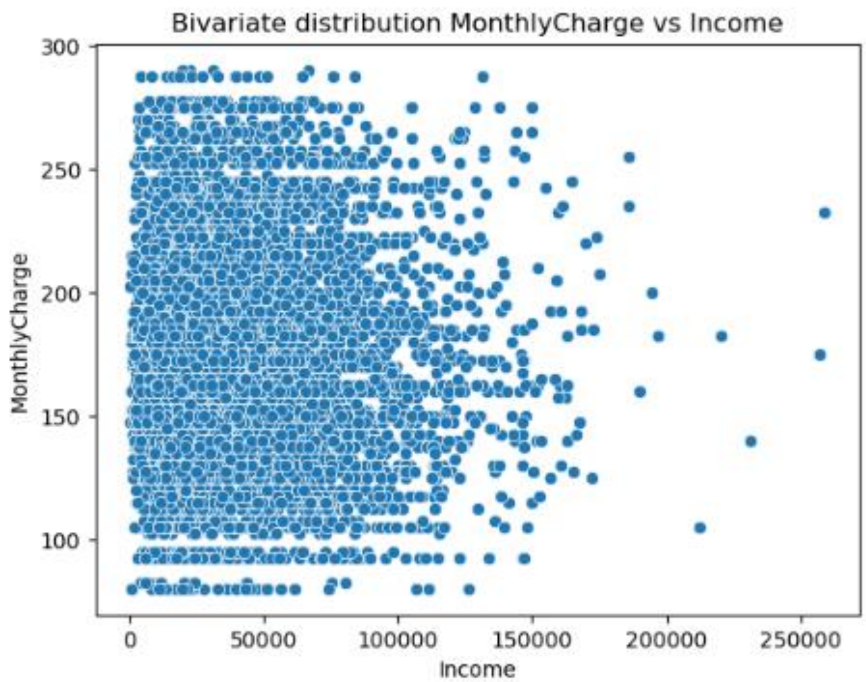
```
# BIVARIATE ANALYSIS
#CONTINOUS VARIABLE 2

print('# BIVARIATE ANALYSIS:CONTINOUS VARIABLE 2')
plt.title('Bivariate distribution MonthlyCharge vs Income')
sns.scatterplot(data=Churn_df, x='Income', y='MonthlyCharge')
plt.show()
print(Churn_df[['MonthlyCharge','Income']].describe())
```

The Distribution

	MonthlyCharge	Income
count	10000.000000	10000.000000
mean	172.624816	39806.926771
std	42.943094	28199.916702
min	79.978860	348.670000
25%	139.979239	19224.717500
50%	167.484700	33170.605000
75%	200.734725	53246.170000
max	290.160419	258900.700000

The Visual

```
# BIVARIATE ANALYSIS:CONTINOUS VARIABLE 2
```



	MonthlyCharge	Income
count	10000.000000	10000.000000
mean	172.624816	39806.926771
std	42.943094	28199.916702
min	79.978860	348.670000
25%	139.979239	19224.717500
50%	167.484700	33170.605000
75%	200.734725	53246.170000
max	290.160419	258900.700000

3. Payment method VS Churn

The code

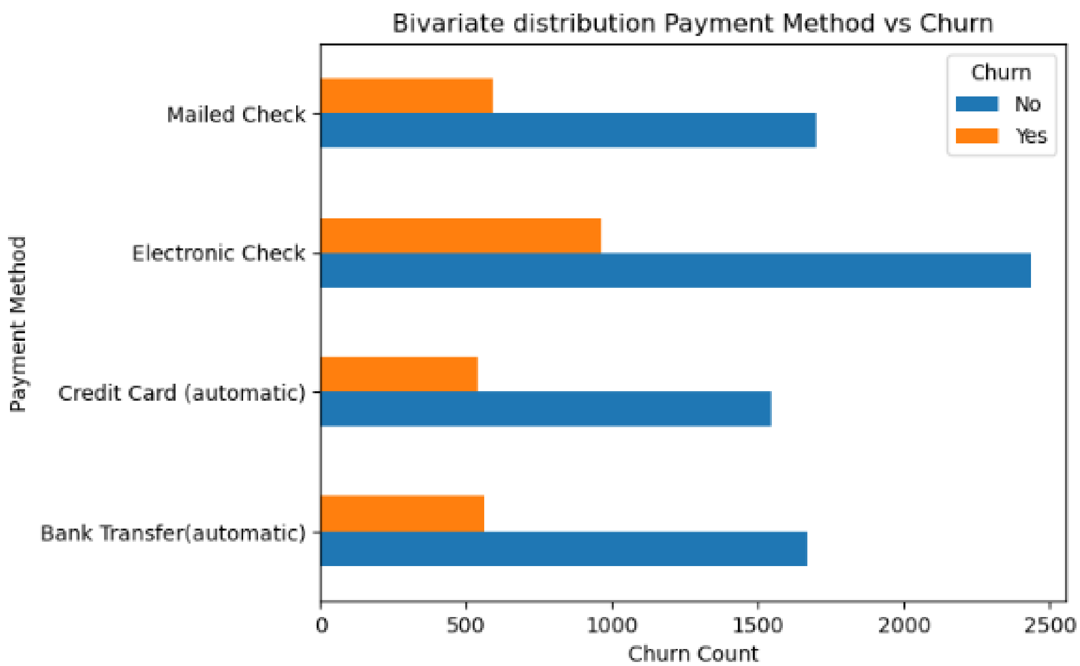
```
# BIVARIATE ANALYSIS
#CATEGORICAL VARIABLE 1
print('# BIVARIATE ANALYSIS:CATEGORICAL VARIABLE 1')
print('')

Count = pd.crosstab(Churn_df['PaymentMethod'], Churn_df['Churn'])
Count.plot(kind='barh', stacked=False)
plt.title('Bivariate distribution Payment Method vs Churn')
plt.ylabel('Payment Method')
plt.xlabel('Churn Count')
plt.legend(title='Churn')
plt.show()
# C Checking Distribution of Churn for the PaymentMethod
print(pd.crosstab(Churn_df['PaymentMethod'], Churn_df['Churn']))
print(Churn_df[['PaymentMethod','Churn']].describe())
```

The distribution

```
Churn                        No   Yes
PaymentMethod
Bank Transfer(automatic)   1671   558
Credit Card (automatic)    1543   540
Electronic Check           2435   963
Mailed Check               1701   589
           PaymentMethod  Churn
count              10000  10000
unique                 4      2
top     Electronic Check     No
freq                3398   7350
```

The Visual

```
# BIVARIATE ANALYSIS:CATEGORICAL VARIABLE 1
```




```
Churn                       No   Yes
PaymentMethod
Bank Transfer(automatic)  1671   558
Credit Card (automatic)   1543   540
Electronic Check          2435   963
Mailed Check              1701   589
           PaymentMethod  Churn
count              10000  10000
unique                 4      2
top     Electronic Check     No
freq                3398   7350
```

4. Area VS Churn

The code

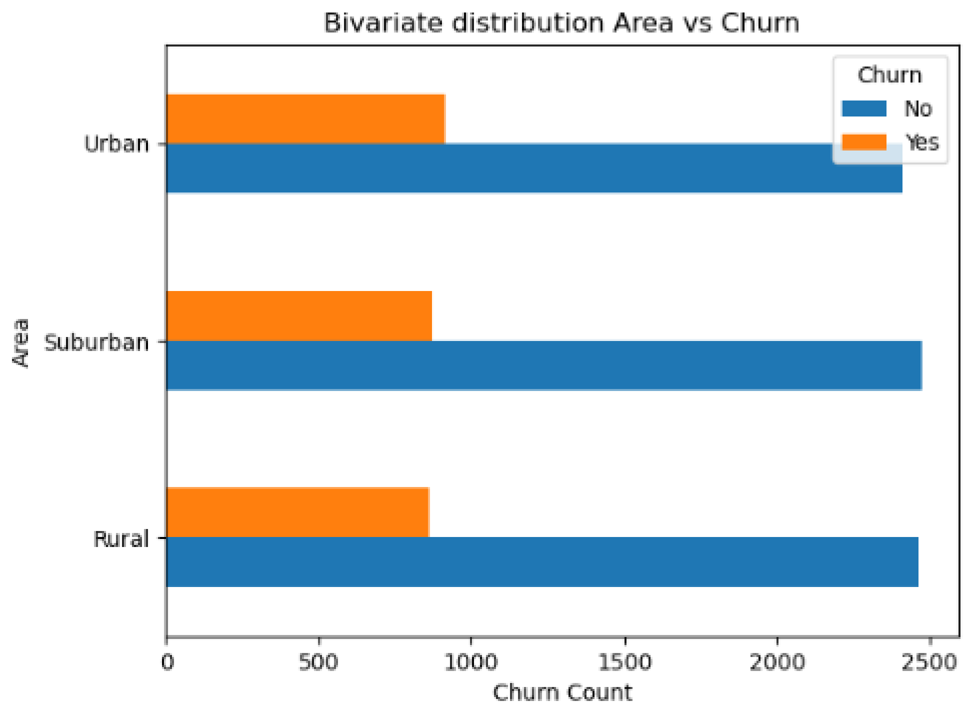
```
# BIVARIATE ANALYSIS
#CATEGORICAL VARIABLE 2
print('# BIVARIATE ANALYSIS:CATEGORICAL VARIABLE 2')
print('')

Count = pd.crosstab(Churn_df['Area'], Churn_df['Churn'])
Count.plot(kind='barh', stacked=False)
plt.title('Bivariate distribution Area vs Churn')
plt.ylabel('Area')
plt.xlabel('Churn Count')
plt.legend(title='Churn')

plt.show()
print(Churn_df[['Area','Churn']].describe())
print(pd.crosstab(Churn_df['Area'], Churn_df['Churn']))
```

The distribution

```
             Area  Churn
count       10000  10000
unique          3      2
top      Suburban     No
freq         3346   7350
Churn       No  Yes
Area
Rural     2464  863
Suburban  2473  873
Urban     2413  914
```

The Visual

```
# BIVARIATE ANALYSIS:CATEGORICAL VARIABLE 2
```




```
              Area   Churn
count        10000   10000
unique           3       2
top       Suburban      No
freq          3346    7350
Churn         No   Yes
Area
Rural       2464   863
Suburban    2473   873
Urban       2413   914
```

E1: RESULTS OF ANALYSIS

With a p-value as large as our output from our chi-square significance testing, pvalue=7.371814027426512e-95, we can reject the null hypothesis that the payment method does not have an impact on the churn rate at a standard significance level of alpha = 0.05 and accept the alternative hypothesis which shows that the payment method does have an impact on the Churn rate.

E2: LIMITATIONS OF ANALYSIS

The analysis was limited due to a lack of understanding of the business data, how the data was collected, and the domain knowledge. More so, the data lacked appropriate dates to explore the effect of payment methods over a time period.

E3:RECOMMENDED COURSE OF ACTION

I recommend further analysis to assess the impact of the payment methods currently utilized in customer churning, looking at various time periods.

F: VIDEO

https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=9fd4979f-9e9e-4c56-9640-b1f900ebc195

G: SOURCES FOR THIRD-PARTY CODE

Vidhya (2024, June 25) 12 Univariate Data Visualizations with Illustrations in Python https://www.analyticsvidhya.com/blog/2020/07/univariate-analysis-visualization-with-illustrations-in-python/#:~:text=Use%20the%20plt.scatter%20%28%29%20function%20of%20matplotlib%20to,as%20the%20index%20of%20the%20data%20frame%20%28df.index%29.

H: SOURCES

McHugh, Mary (2013, January 8) The chi-square test of independence https://pubmed.ncbi.nlm.nih.gov/23894860/

Bose, Shreya (2024 February 28) Python Plotnine: A Beginner Guide to Stunning Data Visualization

https://www.askpython.com/python-modules/python-plotnine-basics

Bobbitt, Zach (2021, November 22) How to Perform Bivariate Analysis in Python (With Examples)

https://www.statology.org/bivariate-analysis-in-python/

Bobbitt, Zach (2021, November 22) How to Perform Univariate Analysis in Python (With Examples)

https://www.statology.org/univariate-analysis-in-python/

Ashwini (2023, November 27) A Quick Guide to Bivariate Analysis in Python

https://www.analyticsvidhya.com/blog/2022/02/a-quick-guide-to-bivariate-analysis-in-python/

I: PROFESSIONAL COMMUNICATION